

Hierarchal planning with JSHOP2

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1 Exercise 1.1: Emergency Service Logistics, Initial Release

In the first exercise, similarly as in PDDL laboratories domain file have been created.

The best performing algorithm in part 1.3 of PDDL project was BFS without any heuristics. In time of 0.77s it managed to find a solution to a problem of size 17. Meanwhile JSOP hierarchal planner managed to solve a problem with size 380.

```
1 (defproblem problem p1
2   (
3     (drone-at drone1 depot)
4     (crate-at crate1 depot)
5     (crate-at crate2 depot)
6     (crate-at crate3 depot)
7     (empty-arm drone1 left)
8     (empty-arm drone1 right)
9     (content-crate food crate1)
10    (content-crate food crate2)
11    (content-crate medicine crate3)
12    (person-at person1 loc1)
13    (person-at person2 loc3)
14    (person-at person3 loc2)
15    (needs-content person2 food)
16    (needs-content person2 medicine)
17    (needs-content person3 food)
18  )
19
20  (
21    (deliver-all)
22  )
23 )

1 (!pick-up-crate crate1 depot drone1 left)
2 (!drone-move depot loc3 drone1)
3 (!drop-crate loc3 person2 crate1 left food drone1)
4 (!drone-move loc3 depot drone1)
5 (!pick-up-crate crate3 depot drone1 right)
```

```

6 (!drone-move depot loc3 drone1)
7 (!drop-crate loc3 person2 crate3 right medicine drone1)
8 (!drone-move loc3 depot drone1)
9 (!pick-up-crate crate2 depot drone1 left)
10 (!drone-move depot loc2 drone1)
11 (!drop-crate loc2 person3 crate2 left food drone1)
12 -----
13
14 Time Used = 0.009

```

Listing 1: First problem and the outcome - three people and three boxes

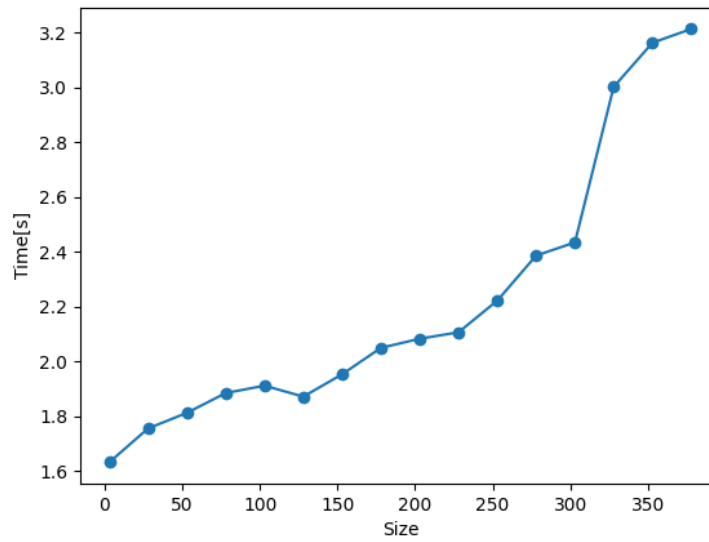


Figure 1: Size of the problem and the time required to find a solution in the tests

It is important to note that time shown on the graph is measured by a python script. Time used measured by JSHOP was below 1 second for all problems.